

JERUSALEM COLLEGE OF ENGINEERING
 (An Autonomous Institution Affiliated to Anna University, Chennai)
Department of Science and Humanities
Academic Year: 2023 - 24
QUESTION BANK
JPH2201 ENGINEERING PHYSICS II

*: K1 - REMEMBER, K2 - UNDERSTAND, K3 - APPLY, K4 - ANALYSE, K5 - EVALUATE, K6 -CREATE.

Unit 1 - SEMICONDUCTING MATERIALS

Course objective - To enlighten the students on the principles of semiconductor materials and electron transport properties

Course Outcome - Students will be able to be familiar with the functioning of semiconductors

PART - A

1	What is semiconductor?	CO1	K1
2	Write the properties of a semiconductor.	CO1	K1
3	Mention any Four advantages of semiconducting materials.	CO1	K1
4	For an intrinsic semiconductor with a band gap of 0.7eV, determine intrinsic carrier concentration at $T = 300\text{K}$ if $m_h^* = 6m_e^*$.	CO1	K3
5	What are the differences between elemental and compound semiconductors?	CO1	K2
6	Compare intrinsic and extrinsic semiconductor.	CO1	K2
7	Write an expression for the concentration of electrons in the conduction band of an intrinsic semiconductor.	CO1	K2
8	Write the expression for the concentration of holes in the valence band of an intrinsic semiconductor.	CO1	K2
9	Write the expressions for carrier concentration in n-type and p-type semiconductor.	CO1	K2
10	Define Fermi level in the case of semiconductors. Mention its position in intrinsic and extrinsic semiconductor at 0K	CO1	K1
11	Write an example for pentavalent and trivalent impurities.	CO1	K1
12	Name any two compound semiconductors with applications.	CO1	K1
13	With increase of temperature the conductivity of a semiconductor increases. Why?	CO1	K1

14	Define Hall-effect and Hall Voltage.	CO1	K1
15	Mention the Uses of Hall effect.	CO1	K1

PART B

1	Derive expressions for the density of electrons and holes in an intrinsic semiconductor. Hence deduce intrinsic carrier concentration.	CO1	K3
2	Obtain an expression for the density of electrons in conduction band in the case of N-type semiconductors.	CO1	K3
3	Obtain an expression for the density of holes in valence band in the case of P-type semiconductors.	CO1	K3
4	Explain the phenomenon of Hall effect. Derive an expression for Hall coefficient for n-type semiconductor and for p-type semiconductor.	CO1	K1

Unit 2 - MAGNETIC MATERIALS

Course objective - To acquaint them with the application of magnetic materials

Course Outcome - Students will be able to assess the magnetic properties of materials

PART - A

1.	Derive the relation between magnetic susceptibility and relative permeability.	CO2	K2
2	What is Bohr magneton?	CO2	K1
3	Give- Curi-Weiss law and its importance.	CO2	K1
4	What is diamagnetism? Mention any four properties of diamagnetic material.	CO2	K1
5	What is paramagnetism? Mention the properties of paramagnetic materials.	CO2	K1
6	What is ferromagnetism? And mention its properties?	CO2	K1
7	What is domain theory of ferromagnetism?	CO2	K1
8	Mention the energies involved in origin of domains in ferromagnetic materials.	CO2	K1
9	What are ferrites? And mention its types.	CO2	K1
10	What are the applications of ferrites?	CO2	K1
11	Draw hysteresis loop and show the retentivity and coercivity in it.	CO2	K2
12	What is hysteresis phenomenon?	CO2	K1
13	What are soft and hard magnetic materials? (or) compare soft and hard magnetic materials on the basis of Hysteresis loop. Give examples.	CO2	K2
14	Mention few magnetic storage devices.	CO2	K1
15	What do you understand by the term, "magnetic domain" and "domain walls"?	CO2	K2

PART B

1	Briefly explain different types of magnetic materials and their properties and applications.	CO2	K2
2	Discuss Weiss theory of ferromagnetism and explain the energies involved during magnetization.	CO2	K1
3	(i) What are Ferro magnetic domains? (ii) Draw a B-H curve for a ferro magnetic material and identify retentivity and coercive field on the curve.	CO2	K1
4	(i) Distinguish between soft and hard magnetic materials. (ii) Give an account of structure, properties and applications of ferrites.	CO2	K2

Unit 3 - SUPERCONDUCTING MATERIALS

Course objective - To impart knowledge on superconducting materials

Course Outcome - Students will be able to make use of superconductors in real life situations

PART - A

1.	What are superconductors? Mention its properties.	CO3	K1
2	Explain the phenomenon of super conductivity.	CO3	K2
3	What is transition temperature?	CO3	K1
4	Explain Meissner effect.	CO3	K2
5	Explain the term critical magnetic field in superconductor.	CO3	K2
6	What are high T _c superconductors? Give examples.	CO3	K1
7	What are the applications of superconductors?	CO3	K1
8	What is isotope effect in superconductivity?	CO3	K1
9	Define magnetic levitation?	CO3	K1
10	What is SQUID?	CO3	K1
11	Super conducting tin has a critical temperature of 3.7K at zero magnetic fields and a critical field of 0.0306 Tesla at 0K. find the critical field at 2K.	CO3	K3

PART - B

1	Explain the properties of Superconductors.	CO3	K2
2	Write an explanatory notes on BCS theory.	CO3	K1
3	Write a note on High T _c superconductors.	CO3	K1
4	Compare Type -I and Type II superconductor (or) Distinguish between soft and hard superconductors.	CO3	K2
5	Describe principal and working of cryotron and SQUID.	CO3	K1

Unit 4 – DIELECTRIC MATERIALS

Course objective – To help them acquire a basic understanding of dielectric materials

Course Outcome – Students will be able to understand the nuances of advanced solid state physics

PART - A

1.	Define dielectric constant.	CO4	K1
2	Define polarization of dielectric material	CO4	K1
3	Mention four type of polarization mechanisms involved in dielectric.	CO4	K1
4	What is electronic polarization?	CO4	K1
5	What is electronic polarization?	CO4	K1
6	What is ionic polarization?	CO4	K1
7	What is orientation polarization?	CO4	K1
8	What is space charge polarization?	CO4	K1
9	What is meant by local field in a dielectric?	CO4	K1
10	Define dielectric loss	CO4	K1
11	What are the factors involved in dielectric loss in a dielectric material?	CO4	K1
12	Define the term dielectric polarizability and susceptibility.	CO4	K1
13	Define dielectric breakdown and dielectric strength.	CO4	K1
14	What are the different types of breakdown mechanisms?	CO4	K1
15	Define intrinsic breakdown.	CO4	K1
16	What is thermal breakdown?	CO4	K1
17	What is chemical and electro chemical breakdown?	CO4	K1
18	What is discharge break down?	CO4	K1
19	What is defect breakdown?	CO4	K1
20	Compare active and passive dielectrics.	CO4	K2
21	Define electronic polarization.	CO4	K1
22	What are the differences between polar and non-polar molecules?	CO4	K1
23	What are the uses of dielectric materials?	CO4	K1
24	What is high-k dielectric?	CO4	K1

PART B

1	Explain the different types of polarization mechanisms involved in a dielectric material.	CO4	K2
2	Derive an expression for electronic and ionic polarisabilities.	CO4	K3
3	Discuss in detail the various dielectric breakdown mechanisms.	CO4	K1
4	Define Internal field obtained the expression for internal field using Lorentz method and hence deduce the Clausius-Mosotti equation.	CO4	K3
5	Discuss the applications of dielectric materials in capacitors and transformers.	CO4	K1
6	Explain dielectric loss and find an expression for dielectric power loss.	CO4	K2

Unit 5 – NANOELECTRONIC MATERIALS

Course objective – To introduce the basic principles in nano electronic devices

Course Outcome – Students will be able to recognize the uses of nano electronic materials

PART - A

1.	Define nano materials.	CO5	K1
2	Define density of states.	CO5	K1
3	Define Fermi energy.	CO5	K1
4	What is quantum size-effect?	CO5	K1
5	What is a quantum confinement?	CO5	K1
6	What is quantum structure?	CO5	K1
7	What are single electron phenomena?	CO5	K1
8	What is single electron tunneling?	CO5	K1
9	What is single electron transistor? Mention its advantages, disadvantages and applications.	CO5	K1
10	What are the limitations of single electron transistor?	CO5	K1

PART B

1	Explain the electron density in bulk material and size dependence of Fermi energy.	CO5	K2
2	Explain quantum confinement and quantum structures in nano material.	CO5	K2
3	Discuss density of states in quantum well, quantum wire and quantum dot structure	CO5	K3
4	Describe construction and working of single electron transistor.	CO5	K1
5	Describe single electron phenomena and single electron transistor	CO5	K1